

Advanced Exercise 2 : Running TFLM Model Inference

Screenshot of Output

```
Running TFLM inference, verification, and cycles
```

```
=== TFLM Functional Verification ===
```

```
Model: ds_cnn_stream_fe.tflite (589000 bytes)
```

```
Tensor arena: 1048576 bytes
```

```
Profile: ds_cnn_stream_fe
```

```
Input profile: label1 board audio input
```

```
Output profile: verification skipped, tolerance=0
```

```
Samples: 1
```

```
--- Sample 1/1 ---
```

```
input bytes=64000 type=1
```

```
Running inference...
```

```
Inference complete.
```

```
Output Data:
```

```
0      : 0xC221B7E7
```

```
1      : 0xC0A9CDE6
```

```
2      : 0x3FA1B7E7
```

```
3      : 0xC131E3E4
```

```
4      : 0xC0E267DD
```

```
5      : 0xC17EB4D9
```

```
6      : 0xC15E5CDE
```

```
7      : 0xC1B3E964
```

```
8      : 0xC139F9E3
```

```
9      : 0xC0815FEC
```

```
10     : 0xC0F293DA
```

```
11     : 0x40523BE0
```

```
Cycles: 9228388741
```

```
Time (500000000 Hz): 184567 ms
```

```
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```

```
TinyRISC-V SoC Platform
```

```
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```

```
l: Lab menu
```

```
t: TFLM inference, verification, and cycles
```

```
p: Performance counter tests
```

```
i: System info
```

```
main> 
```

Design and Implementation

本實作在平台上執行 ds_cnn_stream_fe.tflite 模型，並使用 label1 作為輸入資料。TFLM 呼叫修改後的卷積函式，透過 NPU 的 AXI read 取得 input 與 filter，並使用四路 INT8 SIMD MAC 進行累加。程式在 Invoke() 前後讀取 cycle counter，並在推論完成後印出 12 筆輸出的原始 32-bit 資料。本次推論成功完成，量測結果為 922,838,874 cycles，執行時間為 18,456.7 ms。